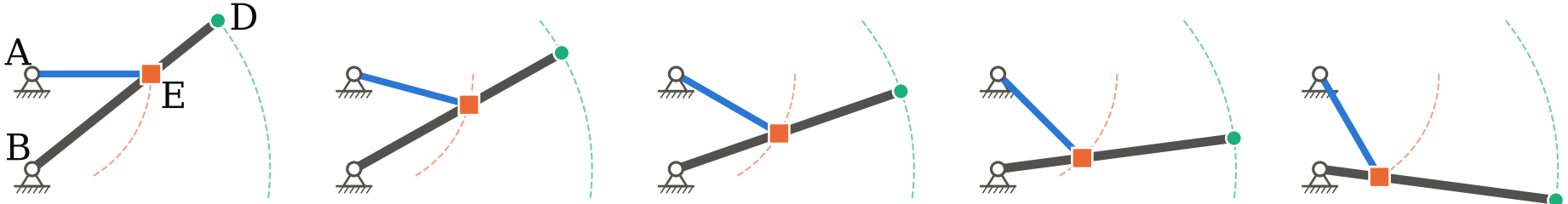$\theta = 0^\circ,\ \alpha = 38.7^\circ$ $\theta = 15^\circ,\ \alpha = 29.3^\circ$ $\theta = 30^\circ,\ \alpha = 19.1^\circ$ $\theta = 45^\circ,\ \alpha = 7.5^\circ$ $\theta = 60^\circ,\ \alpha = -7.5^\circ$